

Linear & Nonlinear Optimization

(P)

$$\min_{x \in \Omega} f(x)$$

where $\Omega \subset \mathbb{R}^n$ (often: $\Omega = \mathbb{R}^n$)
 $f: \Omega \rightarrow \mathbb{R}$

Example 4 (logistic regression)

$$\underline{a}_i \in \mathbb{R}^d$$

$i = 1, \dots, n$

"cats"

$$\underline{b}_i \in \mathbb{R}^d$$

"dogs"

Want to classify a new point as cat or dog.

Define:

$$q(\underline{x}; \underline{y}, \beta) = \frac{\exp(\underline{x}^T \underline{y} + \beta)}{1 + \exp(\underline{x}^T \underline{y} + \beta)}$$

..... probability that $\underline{x}$ is a cat.

input: $\underline{x}$, $\underline{y}$
parameters for optimization: β

We want:

$$q(\underline{a}_i; \underline{y}, \beta) \approx 1 \quad i = 1, \dots, n$$
$$q(\underline{b}_i; \underline{y}, \beta) \approx 0 \quad i = 1, \dots, n$$

Do a maximization:

$$\max_{\underline{y}, \beta} \prod_{i=1}^n q(\underline{a}_i; \underline{y}, \beta) (1 - q(\underline{b}_i; \underline{y}, \beta))$$

Equivalent to maximize the log:

$$\max_{\underline{y}, \beta} \sum_{i=1}^n \left\{ \log \left(\frac{\exp(\underline{a}_i^T \underline{y} + \beta)}{1 + \exp(\underline{a}_i^T \underline{y} + \beta)} \right) + \log \left(1 - \frac{\exp(\underline{b}_i^T \underline{y} + \beta)}{1 + \exp(\underline{b}_i^T \underline{y} + \beta)} \right) \right\}$$

$\left(\frac{1}{\exp(\cdot) + 1} \right)$ $\frac{1}{1 + \exp(\underline{b}_i^T \underline{y} + \beta)}$

$$= \max_{\underline{y}, \beta} \sum_{i=1}^n \left[\log(1 + \exp(-\underline{a}_i^T \underline{y} - \beta)) + \log(1 + \exp(\underline{b}_i^T \underline{y} + \beta)) \right]$$

This is a complicated optimization problem that cannot be solved with paper & pencil!

In the last class, we learned the following first-order necessary condition:

Theorem: (First-order necessary condition) Let $\Omega \subset \mathbb{R}^n$, $f \in C^1(\Omega)$.

If $x^* \in \Omega$ is a local minimum of f on Ω , then for any $d \in \mathbb{R}^n$ that is feasible at x^* , $\nabla f(x^*)^T d \geq 0$.

Theorem (Second-order necessary condition)

Let $\Omega \subset \mathbb{R}^n$, $f \in C^2(\Omega)$. If $x^* \in \Omega$ is a local minimum of f on Ω , then for any feasible direction

(i) $\nabla f(x^*)^T d \geq 0$

(ii) If $\nabla f(x^*)^T d = 0$, then $d^T \nabla^2 f(x^*) d \geq 0$.

Proof: (i) ✓

(ii) say $\nabla f(x^*)^T d = 0$. Again we use Taylor expansion:

$$x(\alpha) = x^* + \alpha d, \quad g(\alpha) = f(x(\alpha)) \quad (\text{as before})$$

know: $g'(\alpha) = \nabla f(x(\alpha))^T d \Rightarrow g'(0) = 0$.

$$g(\alpha) = g(0) + g'(0)\alpha + \frac{g''(0)}{2}\alpha^2 + o(\alpha^2)$$

If $g''(0) < 0$, one can find α small enough such that $g(\alpha) < g(0)$ which contradicts that x^* is local minimum. Hence

$$g''(\alpha) = d^T \nabla^2 f(x^*) d \Rightarrow g''(0) = d^T \nabla^2 f(x^*) d \geq 0. \quad \checkmark$$

Example (cont'): $f(x_1, x_2) = x_1^2 - x_1 + x_2 + x_1 x_2$

$$\nabla f(x^*) = (0, \frac{3}{2})^T$$

$$\nabla f(x^*)^T d = 0 = \frac{3}{2} d_2 = 0 \Leftrightarrow d_2 = 0$$

$$\nabla^2 f(x^*) = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

$$d^T \nabla^2 f(x^*) d = (d_1, 0) \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} d_1 \\ 0 \end{pmatrix} = 2d_1^2 \geq 0 \quad \checkmark$$

Corollary: $\Omega = \mathbb{R}^n$, Then if x^* is a local minimum of f :

(i) $\nabla f(x^*) = 0$

(ii) $\forall d \in \mathbb{R}^n: d^T \nabla^2 f(x^*) d \geq 0$

Def: A symmetric matrix $A \in \mathbb{R}^{n \times n}$ (i.e. $A^T = A$) is positive semidefinite if, for all $\underline{d} \in \mathbb{R}^n$: $\underline{d}^T A \underline{d} \geq 0$ for all $\underline{d}$.

Example: $\min_{x_1, x_2 \geq 0} x_1^2 - x_1 x_2 + 2x_2^2$

Assume soln in interior: $\nabla f(x) = \begin{pmatrix} 3x_1^2 - 2x_1 x_2 \\ -x_1^2 + 4x_2 \end{pmatrix}$

$$\nabla f = 0 \iff \begin{aligned} x_1 &= (9, 0) \\ x_2 &= (6, 9) \end{aligned}$$

$$\nabla^2 f(x) = \begin{bmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{bmatrix}; \quad \nabla^2 f\left(\begin{pmatrix} 6 \\ 9 \end{pmatrix}\right) = \begin{pmatrix} 18 & -12 \\ -12 & 4 \end{pmatrix}$$

is not positive definite as

$$\det(\cdot) = \lambda_1 \lambda_2 < 0$$

$\nearrow \nearrow$
eigenvalues.

$\Rightarrow \begin{pmatrix} 6 \\ 9 \end{pmatrix}$ not a minimum!

$$\text{but } \nabla^2 f\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \Rightarrow \text{local minimum.}$$

Theorem: (Second-order sufficient conditions)

$f \in C^2(\mathbb{R}^n)$, $x^* \in \mathbb{R}^n$ such that:

(i) $\nabla f(x^*) = 0$

(ii) $\nabla^2 f(x^*) > 0$, i.e. positive definite matrix

Then: x^* is a strict local minimum of f .

Proof: (again, Taylor!) $\nabla^2 f(x^*) > 0$. Hence, for all $\underline{d} \in \mathbb{R}^n$:

$$\underline{d}^T \nabla^2 f(x^*) \underline{d} \geq \alpha \|\underline{d}\|^2 \text{ with } \alpha = \lambda_{\min}(\nabla^2 f(x^*))$$

Taylor: $f(x^* + \alpha \underline{d}) - f(x^*) = \alpha \nabla f(x^*)^T \underline{d} + \frac{\alpha^2}{2} \underline{d}^T \nabla^2 f(x^*) \underline{d} + O(\alpha^3)$

$\Rightarrow$ For $\alpha > 0$ small, α^2 dominates the α^3 term, so
 $f(x^* + \alpha \underline{d}) > f(x^*)$ for $\alpha \neq 0$ and any $\underline{d} \in \mathbb{R}^n$. $\square$.

Examples: ① $f(x) = x^2$, $f''(x) = 2$, thus satisfies
SOSC at $\underline{x=0} \Rightarrow x=0$ is strict minimum!

② $f(x) = x^4$, $f''(x) = 12x^2$, thus $f''(0) = 0$
 $\Rightarrow$ does not satisfy SOSC, but is still
strict minimum!

③: $f(x,y) = \frac{1}{2}(x,y) \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - (3,5) \begin{pmatrix} x \\ y \end{pmatrix}$

$$\Rightarrow \nabla^2 f(x,y) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

is positive definite as

$$\det(\dots) = 4 - 1 > 0,$$

$$\text{Tr}(\dots) = 4 > 0.$$

↖ quadratic form,
 $\frac{1}{2} \underline{x}^T A \underline{x} - \underline{b}^T \underline{x} + c$